

ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	0.44	Gender	Male
Height (cm)	170.0	Weight (kg)	70.0
BMI	24.2	Product Info 2	A1
Occupation	Self Employee	Annual Income	\$600,000.00
Insurance Type	Life	Coverage Amount	\$4,500.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 1 of 8
Risk Category	Low Risk
Recommended Annual Premium	\$10,000.00

3. AI Actuarial Analysis

Actuarial Report: Insurance Application Assessment

****Applicant Details:****

* ****Age:**** 0.44 years
* ****Height:**** 170.0 cm
* ****Weight:**** 70.0 kg
* ****BMI:**** 24.2
* ****Product Info 2:**** A1
* ****Occupation:**** Self Employee
* ****Income:**** \$600,000.00
* ****Smoker:**** No (0)
* ****Previous Claims:**** 0
* ****Family History:**** 0
* ****Insurance Type:**** Life
* ****Coverage Amount:**** \$4,500.00
* ****Exercise:**** Active (1)
* ****Alcohol:**** Non-drinker (0)
* ****Gender:**** Male

****Machine Learning Classifier Output:****

* ****Predicted Risk Class:**** 1 (out of 8, lower is better)
* ****Prediction Probability/Confidence:**** 60.3%
* ****Calculated Annual Premium:**** \$10,000.00

1. Risk Assessment

Based on the provided data, a preliminary assessment of the applicant's risk factors presents a significant anomaly that overrides typical actuarial analysis.

****Positive Risk Factors (assuming data consistency):****

* ****Smoking Status:**** Non-smoker (0) is a significant positive factor, reducing health risks.
* ****Previous Claims:**** No previous claims (0) indicates a clean claims history.
* ****Family History:**** No adverse family history (0) is favorable.
* ****Exercise & Alcohol:**** Active lifestyle (1) and non-drinker (0) are positive health indicators.
* ****BMI:**** 24.2 is within the healthy range, indicating no immediate weight-related health concerns.

****Neutral/Potentially Elevated Risk Factors (assuming data consistency):****

* ****Occupation:**** "Self Employee" can sometimes carry slightly elevated income stability risk compared to salaried positions, but does not directly impact mortality risk for Life insurance in most general cases.
* ****Product Info 2:**** "A1" requires further context for risk evaluation.

****Critical Anomaly Affecting All Risk Factors:****

* ****Age Inconsistency:**** The applicant's age is stated as 0.44 years (approximately 5-6 months old). However, other details such as height (170cm), weight (70kg), BMI (24.2), occupation ("Self Employee"), and income

(\$600,000) are characteristic of an adult, not an infant. This severe inconsistency invalidates any risk assessment based on these attributes being simultaneously true. An infant cannot be 170cm tall, weigh 70kg, be self-employed, or earn \$600,000. This indicates a high probability of data entry error, misrepresentation, or potentially fraudulent intent.

2. Comparative Analysis

The current application's ML output (Risk Class 1, Premium \$10,000.00) aligns with two of the top similar historical cases in terms of predicted risk and premium.

* **Application APP-875283:** This case has a very high similarity (98.3%) with the current application, sharing the same Risk Class (1) and Premium (\$10,000.00). However, its status is **rejected**. The high similarity suggests that the underlying applicant profile was nearly identical. The rejection status for such a similar, low-risk (ML-predicted) profile is highly significant and points towards a critical underwriting flag that led to its rejection. Given the age anomaly in the current case, it is plausible that a similar data inconsistency was present and caused the rejection of APP-875283.

* **Application POL-25771:** This case also shares a high similarity (96.1%) with the current application, with the same Risk Class (1) and Premium (\$10,000.00). Crucially, this case was **approved**. This indicates that a low-risk profile with this premium *can* be approved under normal circumstances.

* **Application POL-64336:** This case (96.3% similarity, Risk Class 7, Premium \$30,000.00, Approved) is less relevant for direct comparison due to its significantly higher risk class and premium.

The conflicting outcomes of the highly similar cases (one rejected, one approved) highlight the need for careful scrutiny. The extreme age anomaly in the current application likely places it closer to the "rejected" precedent (APP-875283) due to data integrity issues rather than the "approved" precedent (POL-25771) which presumably had consistent data.

3. Anomaly & Fraud Detection

Several critical anomalies are present:

* **Age-Attribute Discrepancy (Primary Anomaly):** The most severe anomaly is the applicant's stated age (0.44 years) juxtaposed with adult physical attributes (height 170cm, weight 70kg, BMI 24.2) and socio-economic factors (occupation "Self Employee", income \$600,000). This is a physical and logical impossibility and strongly suggests either a severe data entry error or deliberate misrepresentation/fraud.

* **Income vs. Coverage:** An income of \$600,000 is exceptionally high, while the requested coverage amount of \$4,500 is very low for a life insurance policy, especially considering an adult profile. If the applicant were genuinely an adult with this income, the coverage would typically be much higher (e.g., several multiples of income). If the applicant were genuinely an infant, \$4,500 could be a plausible policy amount, but then the income and occupation data would be impossible. This discrepancy further underscores the data inconsistency.

* **Prediction Confidence:** A confidence score of 60.3% is relatively low for an ML classifier, especially for a Risk Class 1 prediction. While not an anomaly itself, it suggests the model may have encountered some unusual input combinations or is not highly certain, which further justifies manual review given the glaring data issues.

4. Premium Recommendation

The ML classifier calculated an annual premium of \$10,000.00 for a predicted Risk Class 1. This premium is consistent with historical approved and rejected cases (APP-875283 and POL-25771) that were also classified as Risk Class 1.

* If the applicant's data were accurate for a healthy adult (i.e., ignoring the age and assuming it was a typo for an adult, perhaps 44 years old), a \$10,000.00 premium for a \$4,500 coverage could be considered extremely high, suggesting the coverage amount might be in error, or the premium calculation assumes a much higher coverage or different product.

* However, given the ****severe data inconsistencies****, the calculated premium, while mechanically derived from the ML model, cannot be confidently recommended or justified. The underlying data for the calculation is fundamentally flawed, rendering the premium amount unreliable for the true risk.

Therefore, the \$10,000.00 premium cannot be justified or validated without resolving the primary data anomalies.

5. Underwriting Decision Recommendation

****Recommendation: Refer for Manual Review (Urgent Data Validation Required)****

Given the severe and contradictory data points, particularly the age-attribute mismatch (0.44-year-old infant with adult physical characteristics, income, and occupation), it is imperative to refer this application for immediate and thorough manual review.

****Rationale:****

1. ****Data Integrity:**** The application contains fundamental inconsistencies that invalidate the machine learning output and any standard actuarial assessment.
2. ****Potential Error/Fraud:**** The discrepancies strongly suggest either a critical data entry error or an attempt at misrepresentation/fraud.
3. ****Unreliable ML Output:**** While the ML classifier provided a risk class and premium, its output is based on flawed input data and cannot be trusted. The relatively low confidence score (60.3%) also supports caution.
4. ****Precedent of Rejection:**** A highly similar historical case (APP-875283) was rejected, likely due to such underlying issues.

****Required Actions for Manual Review:****

- * Verify the applicant's true age, height, weight, occupation, and income with supporting documentation.
- * Clarify the intended coverage amount and insurance product.
- * Investigate the source of the data inconsistency.

No underwriting decision (approve or reject) should be made before these critical data points are validated and reconciled.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **\$10,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.